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Influence of sea surface waves on numerical modeling of an oil spill: Revisit of symphony wheel accident

Weizeng Shao [a,*], Jiale Chen [a], Song Hu [a], Yiqiu Yang [a,b], Xingwei Jiang [c], Wei Shen [d], Huan Li [e]

[a] College of Oceanography and Ecological Science, Shanghai Ocean University, Shanghai, China
[b] Key Laboratory of Marine Hazards Forecasting, National Marine Environmental Forecasting Center, Ministry of Natural Resources, Beijing, China
[c] National Satellite Ocean Application Service, Ministry of Natural Resources, Beijing, China
[d] China Waterborne Transport Research Institute, Ministry of Transport, Beijing 100088, China
[e] National Marine Data and Information Service, Ministry of Natural Resources, Tianjin, China

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ABSTRACT

The greatest purpose of this study is to analyze the importance of surface waves on the hindcasting of the oil spill through the Symphony wheel accident in the Qingdao coastal waters. During the accident period, a total of four synthetic aperture radar (SAR) images by Gaofen-3 (GF-3) were acquired from 2 to 19 May 2021. The hindcasting of two sea surface dynamics, namely currents and waves, is carried out using a coupled marine numeric model. This model, known as the finite-volume community ocean model-simulating waves nearshore (FVCOM-SWAVE), employs a triangular grid. Simulated significant wave height (SWH) is validated against remotely sensed product by the Haiyang-2B (HY-2B) altimeter on April 2021 yields a root mean square error (RMSE) of 0.38, a correlation coefficient (COR) of 0.78, and a scatter index (SI) of 0.34. Subsequently, Stokes drift estimated by waves are included to hindcasting oil spills using the oil particle-tracing method. The bias of the spatial coverage (SAR minus simulations) of an algorithm called the constant false alarm rate (CFAR) is −73.92 km^2 with Stokes drift, which is significantly less than the 55.45 km^2 coverage without Stokes drift. Moreover, compared with model-simulated oil spills, the bias of the geographic location at the center point with Stokes drift is 8.18 km, which is less than the 12.95 km bias without Stokes drift. These results demonstrate that Stokes drift needs to be included in the prediction of oil spills.

1. Introduction

Among various types of marine pollution, oil spills are an important aspect of marine environmental monitoring, and they pose serious hazards to the marine environment, marine organisms, and economics. Remote sensing techniques have advantages such as near real-time, large swath coverage, and fine spatial resolutions. Therefore, multi-source satellite data play an essential role in handling oil spills. Although oil spills are intuitive on the optical image (Kolokoussis and Karathanassi, 2018) and along-track scanning radiometer (Niclòs et al., 2014), oil spills are undetectable under cloudy weather conditions. Furthermore, as revealed in a previous study (Feng and Hu, 2016), the valid data for sea surface monitoring only account for 5% of all optical images.

Compared with other remote sensing sensor operating microwave frequencies (Chiu et al., 2018), the distinctive feature of synthetic aperture radar (SAR) is superior spatial resolution and wide swath coverage, such as the 1 m pixel size acquired in spotlight mode for the X-band TerraSAR-X (Shao et al., 2014). This apparent advantage leads to the development of SAR dynamic sea surface retrieval, i.e., winds (Yao et al., 2022; Hu et al., 2023), waves (Schulz-Stellenfleth et al., 2005), and currents (Shao et al., 2024), and target detection, i.e., ships (Zhang et al., 2022) and oil spills (Meng et al., 2024; Jiang et al., 2023). SAR receives resonance signal backscattering with the sea surface gravity waves (Lyzenga, 1986). As there is an oil film, the sea surface tension and roughness change, making the sea surface smoother than that without an oil film (Brekke et al., 2014). Due to the mirror reflection associated with smooth seawater surfaces, the backscattering signals received by SAR are reduced, resulting in black patches or strip-shaped features on the images (Meng et al., 2022). In the literature, oil spill detection on a SAR image only includes distinguishing between an oil slick and seawater. The well-known algorithm, the constant false alarm

* Corresponding author.
E-mail address: wzshao@shou.edu.cn (W. Shao).

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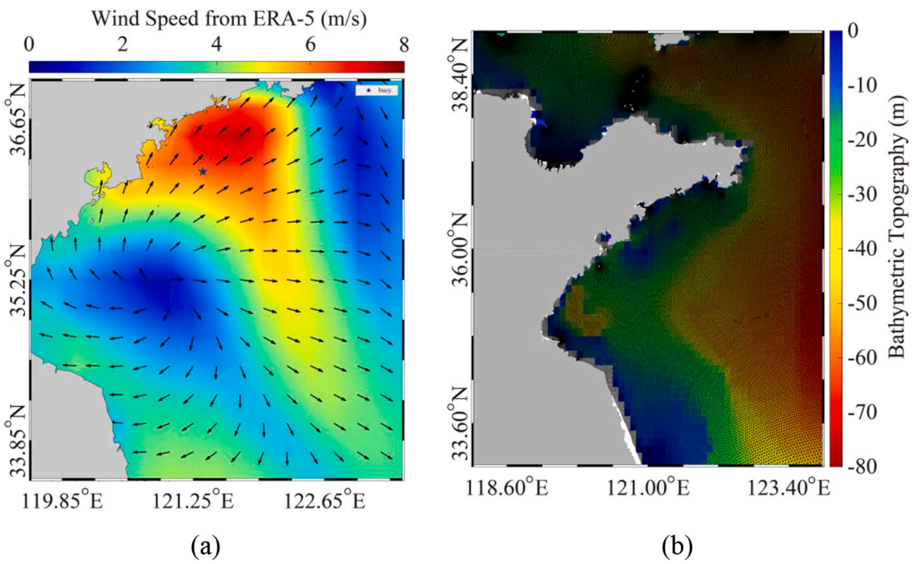

Fig. 1. (a) European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis (ERA-5) wind map that was acquired at 09:00 UTC on 2 May 2021, in which the blue star represents the geographic location of the buoy. (b) The unstructured grid of the computational domain in the finite-volume community ocean model (FVCOM)-simulating waves nearshore (SWAN) (FVCOM-SWAVE) model overlaid the water depth from the General Bathymetric Chart of the Oceans (GEBCO). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

rate (CFAR), can be conveniently applied for oil spill detection utilizing co-polarized (vertical–vertical (VV) and horizontal–horizontal (HH)) SAR measurements (Wang et al., 2015a, 2015b). Taking advantage of amplitude coherence and the co-polarized phase difference (CPD) standard deviation (Buono et al., 2019) estimated from a dual-polarized image, i.e., the slick-free to slick-covered polarization ratio (Skrunes et al., 2014), can improve the accuracy of SAR oil spill detection (Velotto et al., 2014). In addition, compact polarimetric SAR measurements (Zhang et al., 2017) showcase the superiority of oil spill monitoring due to the optimal selection between polarization and resolution. In the era of artificial intelligence, deep learning has been implemented for large-scale detection of oil spills from SAR images (Bianchi et al., 2020). However, the distortion caused by oil look-alikes, i.e., low wind streaks and upwelling, reduces the applicability of the mentioned methods.

Although SAR has an excellent ability to detect real-time oil spills, the diffusion trend of an oil spill on the sea surface is predictable using a numerical model. Specifically, this is confirmed in the revisit of the Sanchi accident (Li et al., 2019; Qiao et al., 2019; Yang et al., 2021a). In the past two decades, many numerical models targeted for predicting oil slick in near-real time have been developed. State-of-the-art oil slick prediction physics have been comprehensively discussed in several reviews (Fingas, 1995; Afenyo et al., 2016), i.e., Lagrangian transport, Monte Carlo stochastic calculation, and the particle-tracing method. The basis of the oil particle-tracing method is the common assumption that large-scale oil spills can be idealized using a large number of small particles (Garcia-Martinez and Flores-Tovar, 1999). Therefore, utilizing particle-tracing method is able to directly simulate the actual oil diffusion under complicated ocean environments, based on which, the oil spill prediction model has been widely used (Guo and Wang, 2009). In fact, air-sea interface dynamics, i.e., winds, currents, and waves, determine oil slick diffusion and transport. In this sense, the accuracy achieved by the oil particle-tracing method relies on prior knowledge of the sea surface dynamics. In the recent study, it was found the inclusion of Stokes can improve the oil spill trajectory (Lee et al., 2020; Yang et al., 2021b). However, it is worthy to conduct the quantitative evaluation of the Stokes effect on the oil spill prediction.

At present, numerical models are popularly used for hindcasting and predicting sea surface dynamics, i.e., the finite-volume community ocean model (FVCOM) (Chen et al., 2003) and hybrid coordinate ocean model (HYCOM) (Backeberg et al., 2009) for circulation and WAVEWATCH-III (WW3) (Hsiao et al., 2020) and the simulating waves nearshore (SWAN) (Rogers et al., 2003) models for waves. Among these models, FVCOM and SWAN share a unique triangular grid that meets the requirements in coastal waters (Zhao et al., 2023). In the direction of wave propagation, the movement of water particle is not closed, resulting in minor displacement of the water particle. This is defined as Stokes drift (Herbers and Janssen, 2016), in particular, the order of the Stokes transport is proportional to 50% of Ekman transport (Wang et al., 2015a, 2015b). Moreover, as mentioned in previous studies (Sun et al., 2021; Yao et al., 2023), wave-induced effects have nonnegligible influence on the sea surface temperature in global oceans and during tropical cyclones.

Panama-flagged general cargo ship ‘Sea Justic’ traveling from Port Sudan to Qingdao collided with the anchored Liberian oil tanker namely ‘A Symphony’ at coastal waters of Qingdao on April 27, 2021. The collision resulted in a spill of approximately 9400 tons of oil into the sea and the Symphony wheel accident occurred about 40 nautical miles (approximately 75 km) from Qingdao Port at (35.73°N,120.97°E) in the Yellow Sea. The main purpose of our work is to quantitative analyze of the impact of Stokes drift on oil spill simulation with respect to SAR images. In this study, a model that couples the FVCOM and SWAN numerical models, named the FVCOM-SWAVE model (Qi et al., 2009), is employed to synchronously simulate the sea surface currents and waves during the Symphony wheel accident in the Qingdao coastal waters. Then the European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis (ERA-5) wind and simulated current fields are treated as the forcing fields in the oil particle-tracing method. In particular, the results of hindcasting of oil spills with and without Stokes drift estimated from the simulated waves are established through the retrieval results from four GF-3 images. The rest of this paper is structured as follows: Section 2 presents the available datasets, i.e., ERA-5 wind, HYCOM fields, GF-3 images, and remotely sensed products from the Haiyang-2B (HY-2B) altimeter. The methodology, i.e., the settings of the FVCOM-SWAVE and the oil particle-tracing method and algorithm CFAR, is introduced in Section 3. Section 4 presents the sensitivity experiment conducted to investigate the role of sea surface waves on the numerical modeling of oil spills. At last, the main conclusions are briefly given in Section 5.

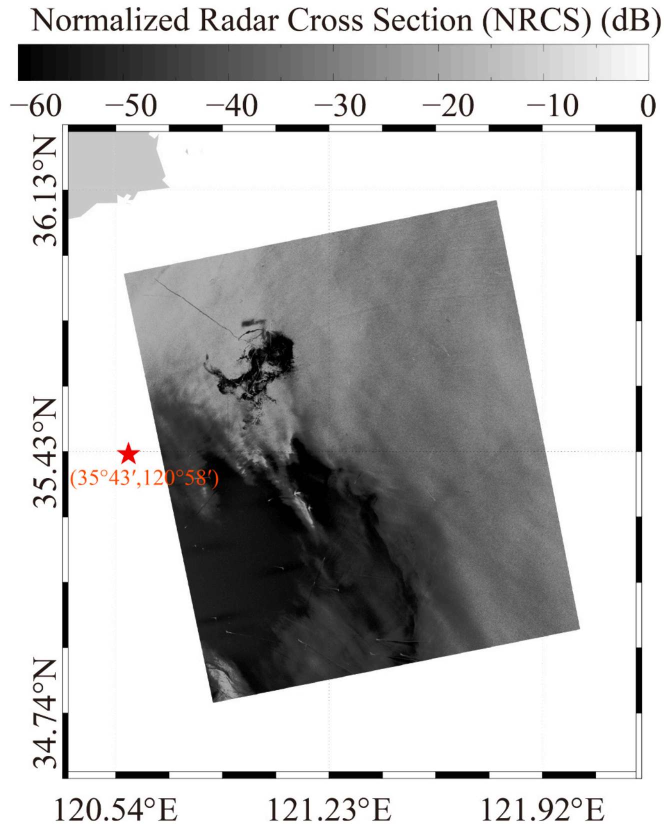

Fig. 2. Normalized radar cross section (NRCS) map of an GF3 image in vertical–vertical (VV) polarized that was acquired at 09:50 UTC on 2 May 2021. Noted that the star denotes the geographic location of the Symphony wheel.

2. Dataset

The datasets mainly consist of two categories: open-accessed sources for hindcasting of sea surface dynamics and oil spills, i.e., the ERA-5 wind and HYCOM fields, and GF-3 images and the HY-2B product.

2.1. ERA-5 wind and HYCOM fields

Since 1979, the ECMWF has continuously released an atmospheric-marine dataset for use by the public. In particular, ERA-5, which is a reanalysis assimilated using measurements from moored buoys and remote sensing products (Balmaseda et al., 2013), is recognized by the scientific community as a valuable source. Here, the ERA-5 wind field at a 0.25° grid of spatial resolution and 1 h of temporal resolution is directly employed as the forcing field of the FVCOM-SWAVE model and the oil particle-tracing method. As an example, Fig. 1a shows the ERA-5 wind map at 09:00 UTC on May 2, 2021, in which the blue star represents the location of an available buoy.

In our previous studies, it was confirmed that considering wave–current interaction can improve the applicability of wave simulations by using a numeric wave model (i.e., SWAN model (Yang et al., 2020) and WW3 model (Hu et al., 2020)). Here, the coupled FVCOM-SWAVE is used to synchronously simulate currents and waves sharing a unique grid. The unstructured grid for the computational region is illustrated in Fig. 1b, in which the land boundary and island boundaries gained from the General Bathymetric Chart of the Oceans (GEBCO) are densified. The best spatial resolution in the horizontal direction is 1 km, including 30,228 computational nodes and 253,903 triangular grids. The TPXO.5 tide data and four fields from HYCOM with 0.08 grid and 3 h spatial and temporal resolutions, i.e., the sea surface temperature salinity, sea current, and water level, are set as the open boundary conditions, ensuring that the momentum and water mass inside and outside the computational domain can be exchanged.

Table 1
Information about the Gaofen-3 (GF-3) images in vertical–vertical (VV) polarization.

Imaging Time	Range × Azimuth Resolution (m)	Incidence Angle (°)	Range × Swath (km)
2021-05-02 09:50 UTC	10 × 10	24.09–31.79	110 × 110
2021-05-04 10:07 UTC	25 × 25	42.75–47.82	130 × 130
2021-05-09 09:58 UTC	25 × 25	36.13–43.29	130 × 130
2021-05-19 09:47 UTC	25 × 25	15.28–26.60	130 × 130

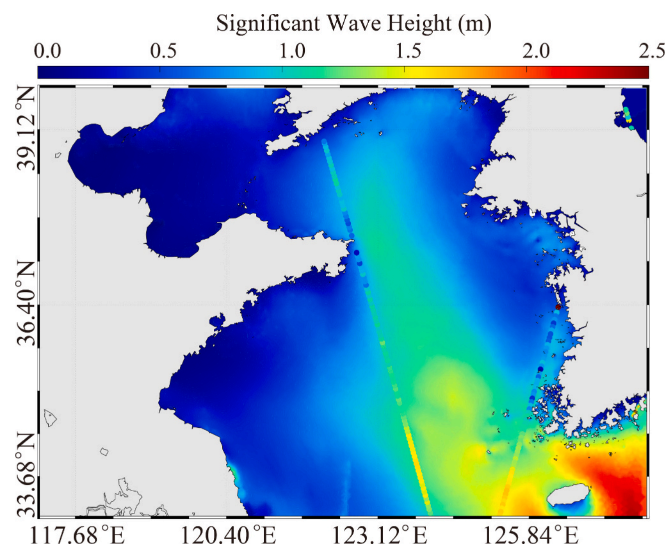

Fig. 3. The swath-gridded significant wave height (SWH) map from the Haiyang-2B (HY-2B) altimeter on 2–3 May 2021 overlain with the simulated SWH from the FVCOM-SWAVE model at 09:00 UTC on 2 May 2021.

2.2. SAR images and HY-2B product

In total, four VV-polarized GF-3 images acquired in fine stripmap (FS) and stander stripmap (SS) modes were acquired during the Symphony wheel accident. These images have been processed to a Level-1 product, and the calibration method has been described by Shao et al. (2023). The method for calibrating the GF-3 image is described by following equation:

$$\sigma_0 = DN^2\left(\frac{M}{32767}\right) - N \tag{1}$$

in which σ_0 is the NRCS united in dB that dependent on polarization; *DN* the intensity; and *M* and *N* are constants annotated with SAR raw data. It is necessary to figured out that historical GF-3 image has been recalibrated since November 2019 so as to improve the imaging quality of image. Fig. 2 presents a normalized radar cross section (NRCS) map of GF3 image, in which the star denotes the geographic location of the Symphony wheel. Detailed information about all of the images is presented in Table 1.

The altimeter can obtain the accurate sea level height (SLH) and significant wave height (SWH) via inversion of the SLH (Zhang et al., 2003). It has been revealed in a recent work (Shao et al., 2021) that operational geophysical data records (GDRs) of the SWH measured by the HY-2B altimeter are quite reliable. To study the applicability of simulations conducted by the FVCOM-SWAVE, the GDR products from the HY-2B altimeter on April 2021 were collected. Fig. 3 depicts the swath-gridded SWH map from the HY-2B altimeter on 2–3 May 2021 overlain with the simulated SWH from the FVCOM-SWAVE model at 09:00 UTC on 2 May 2021. It should be noted that the diagonal line represents the track of the HY-2B altimeter on 2–3 May 2021.

3. Methodology

In this section, the FVCOM-SWAVE model and calculation of Stokes drift are presented. Then, the oil particle-tracing method for hindcasting oil spill diffusion is introduced. Finally, the CFAR technique for oil spill detection from VV-polarized SAR images is presented.

3.1. FVCOM-SWAVE

The SWAN model has been widely used in offshore wave modeling studies (Sun et al., 2018). It was initially designed for use with a structured grid, and recently the unstructured grid method has been added as a model option. However, the SWAN model can only compute waves, and it cannot fully capture the coupling of the waves and currents, such as tidal currents near shores. Qi et al. (2009) successfully adapted the SWAN model and embedded it into the FVCOM to produce the FVCOM-SWAVE model. This model is capable of fitting complex shoreline geometries, making it particularly suitable for simulating wave–current coupling under complex coastal terrain conditions along shorelines. It has been proven that the coupled FVCOM-SWAVE is able to robustly capture the wave characteristics in both deep and shallow regions at the east U.S. coastal waters by fully considering the complex shoreline geometries and high-resolution water depth distribution (Qi et al., 2009).

In this study, we used the FVCOM-SWAVE model to simulate hydrodynamics, including waves and currents. The complete equation of the wave spectra is shown as follows:

$$\frac{\partial \mathrm{N}}{\partial \mathrm{t}}+\nabla\cdot\left[\left(\vec{\mathrm{C}}_{\mathrm{g}}+\vec{\mathrm{V}}\right)\mathrm{N}\right]+\frac{\partial \mathrm{C}_{\sigma}\mathrm{N}}{\partial\sigma}+\frac{\partial \mathrm{C}_{\theta}\mathrm{N}}{\partial\theta}=\frac{\mathrm{F}}{\sigma} \tag{2}$$

where t is the time; N is the density of wave action spectral; C_g is the group velocity and V is the current velocity; C_σ and C_θ are the wave propagation velocities in term of frequency and direction, respectively; σ and θ represents the relative frequency and the wave direction, respectively; and the sum of the source and sink is F associated with the input of wind and dissipations, i.e., wave breaking, bottom friction and multiply wave-wave interactions.

Like the algorithms in the SWAN model (Hsu et al., 2005), the FVCOM-SWAVE model splits the wave action density equation into four equations:

$$\frac{\mathrm{N}^{\mathrm{n}+\frac{1}{4}}-\mathrm{N}^{\mathrm{n}}}{\Delta \mathrm{t}}+\frac{\partial(\mathrm{C}_{\sigma}\mathrm{N})}{\partial\sigma}=0 \tag{3}$$

$$\frac{\mathrm{N}^{\mathrm{n}+\frac{2}{4}}-\mathrm{N}^{\mathrm{n}+\frac{1}{4}}}{\Delta \mathrm{t}}+\frac{\partial(\mathrm{C}_{\theta}\mathrm{N})}{\partial\theta}=0 \tag{4}$$

$$\frac{\mathrm{N}^{\mathrm{n}+\frac{3}{4}}-\mathrm{N}^{\mathrm{n}+\frac{2}{4}}}{\Delta \mathrm{t}}+\nabla\cdot\left[\left(\vec{\mathrm{C}}_{\mathrm{g}}+\vec{\mathrm{V}}\right)\mathrm{N}\right]=0 \tag{5}$$

$$\frac{\mathrm{N}^{\mathrm{n}+1}-\mathrm{N}^{\mathrm{n}+\frac{3}{4}}}{\Delta \mathrm{t}}=\frac{\mathrm{S}}{\sigma} \tag{6}$$

where n is the n^{th} time step and Δt is the time step. Eq. (3) follows the principle of flux-corrected transport method; Eq. (4) is calculated using the Crank–Nicolson method; and Eq. (6) is solved using the same scheme as the WW3 model (Qi et al., 2009). It should be noted that Eq. (5) is revised using the finite-volume second-order upwind scheme used in the FVCOM model.

Table 2
Settings of FVCOM-SWAVE model.

Initial fields	sea surface temperature, salinity, sea current and water level from HYCOM with 0.08 grid and 3 h spatial and temporal resolutions，ERA-5 wind field(0.25° grid of spatial resolution and 1 h of temporal resolution)TPXO.5 tide data;
Water depth	General Bathymetric Chart of the Oceans (GEBCO)
Study area range	31.4–40.9°N, 117.6–127.9°E;
Outputs	Current, sea temperature, water level, salinity, wave, significant wave height, significant wave period;
Timing resolution of outputs	The time resolution is 0.5 h, and the finest grid resolution is 0.1 km；
Validation sources	Haiyang-2B (HY-2B) altimeter and the measurements from a buoy

Eq. (5) is integrated into the control volume around each node to obtain the following equation:

$$N^{n+\frac{3}{4}}=N^{n+\frac{2}{4}}-\frac{\Delta t}{\Omega}\oint_{l}C_nN_l dl \tag{7}$$

where Ω is the central region, l is the edge of the control region, and C_n is the component of $\vec{C}_g+\vec{V}$ normal to l. The algorithm of the action density N_l depends on the sign of C_n. A detailed description of the FVCOM-SWAVE model has been provided by Qi et al. (2009). The settings of FVCOM-SWAVE model is presented in the Table 2. The Stokes drift U_0 can be estimated as a function of the SWH H_s and mean absolute wave period T (TM01) as follows:

$$U_0=\frac{2\pi^3H_s^2}{gT^3}D \tag{8}$$

where g is the constant of gravity (=9.8 m^2/s) and D is the wave propagation direction.

3.2. Oil particle-tracing method

The oil parcel concept regards the spilled oil as consisting of many droplets of oil. Therefore, the oil film can be discretized and tracked in a Lagrangian frame during oil spill modeling (Maderich et al., 2012). Since 2007, China's National Marine Environmental Forecasting Center (NMEFC) has independently developed its own oil spill model, named National Operational Oilspill Forecasting Model (NOOFM) (Yang et al., 2017; Yang et al., 2021b). In this paper, Symphony wheel accident is a main object of study. It is a ship oil spill on the sea, and the spilled oil is mainly concentrated on the sea surface that can be detected by satellite. Therefore, a 2-dimensional oil spill model would characterize the movement of oil spill on sea surface in this accident.

Virtual particles are employed during numerical oil tracking. Each droplet of oil carries the equal masses, which depend on the number of droplets and quality of spilled oil. The drift of oil on the sea is influence by sea surface currents, wind, waves and turbulent diffusion. The following formula can be used to describe the drift velocity of an oil droplet:

$$\mathrm{u}_0=\mathrm{u}_\mathrm{c}+\alpha(\mathrm{u}_\mathrm{a}\cos\beta-\mathrm{v}_\mathrm{a}\sin\beta)+\mathrm{u}_\mathrm{w}+\mathrm{u}' \tag{9}$$

$$\mathrm{v}_0=\mathrm{v}_\mathrm{c}+\alpha(\mathrm{u}_\mathrm{a}\sin\beta+\mathrm{v}_\mathrm{a}\cos\beta)+\mathrm{v}_\mathrm{w}+\mathrm{v}' \tag{10}$$

where u_0 and v_0 are the horizontal velocities of oil droplet in longitudinal and latitudinal directions; u_c and v_c are the surface ocean current velocities in eastern and northern directions, respectively; u_a and v_a come from wind velocities of 10 m above the surface of sea; α and β are factors related to wind, which named wind drift factor and drift angle. The recommended values are 0.02 and 0° in this study (Li et al., 2013).

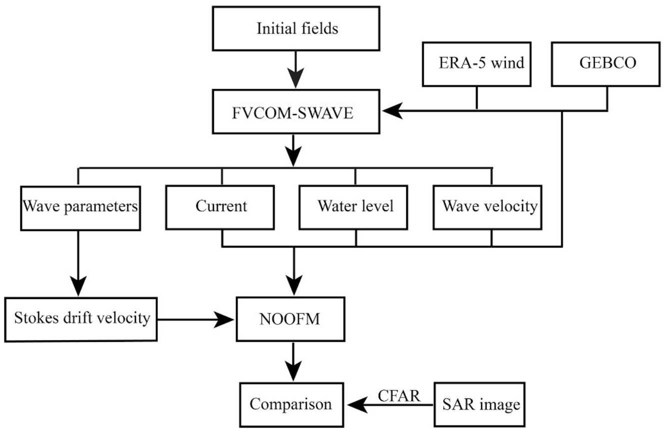

Fig. 4. The flowchart of the methodology in this study.

The terms of the velocities of the Stokes drift u_w and v_w in the x and y directions are tested in the sensitivity experiment. The turbulent diffusions $u^{'}$ and $v^{'}$ are expressed as follows:

$$\begin{cases} u^{'} = (2R - 1)\sqrt{c^{'}E_x\Delta t} \\ v^{'} = (2R - 1)\sqrt{c^{'}E_y\Delta t} \end{cases} \tag{11}$$

in which E_x and E_y are the turbulent coefficients in x and y direction; R is a random number between 0 and 1; $c^{'}$ is the scaling coefficient; Δt is the time integration step in oil spill model.

3.3. CFAR

The four key presumptions of the CFAR method were shown as below: (1) Any sub-scene within a pixel dimension will not significantly impact other sub-scene; (2) the phase of sub-scene component follows a uniform distribution within the range of $[-\pi, \pi]$; (3) The phase random variables for each sub-scene are independent, and naturally, some interconnected sub-scene collectively creates a sub-scene center; (4) the amplitude random variable and the phase random variable of each sub-scene are uncorrelated.

For the SAR backscatter signal that follows the previously mentioned four assumptions, the SAR ocean imagery conforms to a Gaussian distribution:

$$f_i(x) = \frac{1}{\sqrt{2\sigma_i}} exp\left(-\frac{(x-\mu_i)^2}{2\sigma_i{}^2}\right) \tag{12}$$

where $f_i(x)$ represents the sub-scene i probability density function; with the average value μ_i acting as the shape argument. The standard deviation σ_i, which serves as the characteristic of data distribution in each sub-scene. It measures the dispersion of the image brightness and defines the fundamental attributes of the Gaussian shape and density function $F(k)$.

The flowchart of oil detection by the CFAR method is summarized as below: firstly, choosing a specific sub-scene, for instance, 256 × 256 pixels, surrounding a given pixel x_c to serve as the reference window in accordance with the principle outlined in Eq. (13); and secondly, setting a threshold x_0 based on the statistical properties of the reference window, ensuring that the detection process with a false alarm rate P_f linked to the threshold x_{0i} of the local area i is elucidated in Eq. (14):

$$\begin{cases} x_c \text{ is the target signal, if } x_c > x_0 \\ x_c \text{ is the clutter signal, otherwise} \end{cases} \tag{13}$$

$$x_{0i} = \mu i + \sqrt{-2\sigma_i^2 ln(P_f)} \tag{14}$$

where, the x_{0i} is the threshold; μi represent the image mean. In this study, the μi and variance σ_i were calculated from the sub-scenes.

The flowchart of the methodology in this study is presented in Fig. 4.

4. Results and discussion

In this section, the hindcasting results obtained using the FVCOM-SWAVE model are validated against remotely sensed products from the HY-2B altimeter. Utilizing the oil spills detected using several GF-3 images, the effect of Stokes drift produced by wave on oil spill

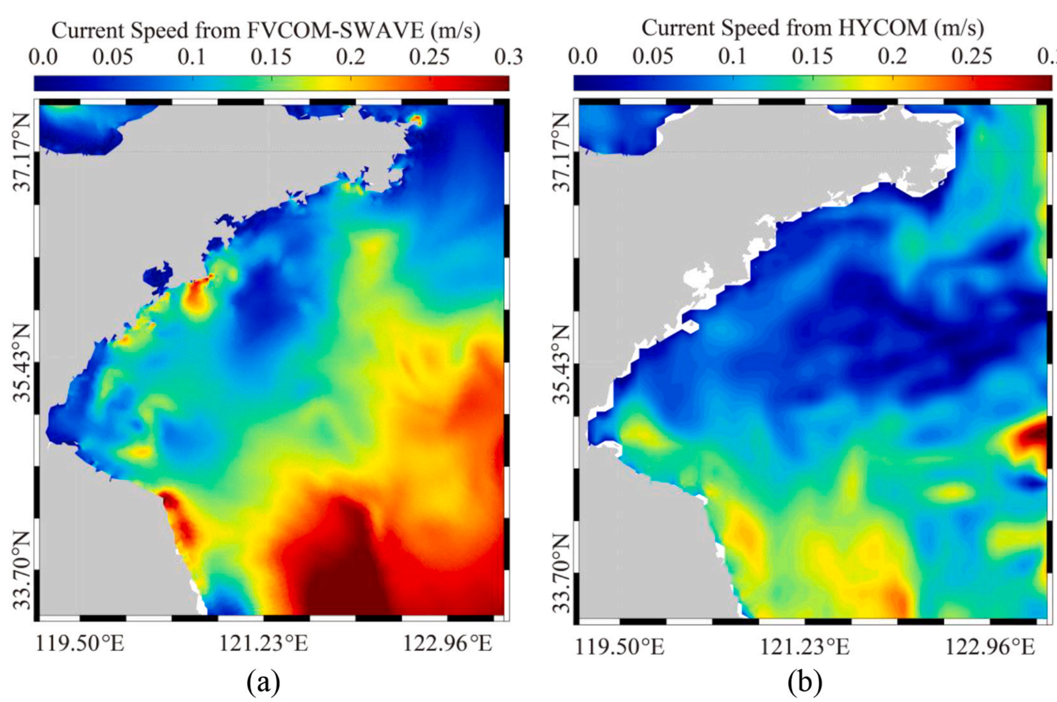

Fig. 5. Maps of the current at 12:00 UTC on 3 May 2021 from two datasets: (a) FVCOM-SWAVE and (b) HYCOM.

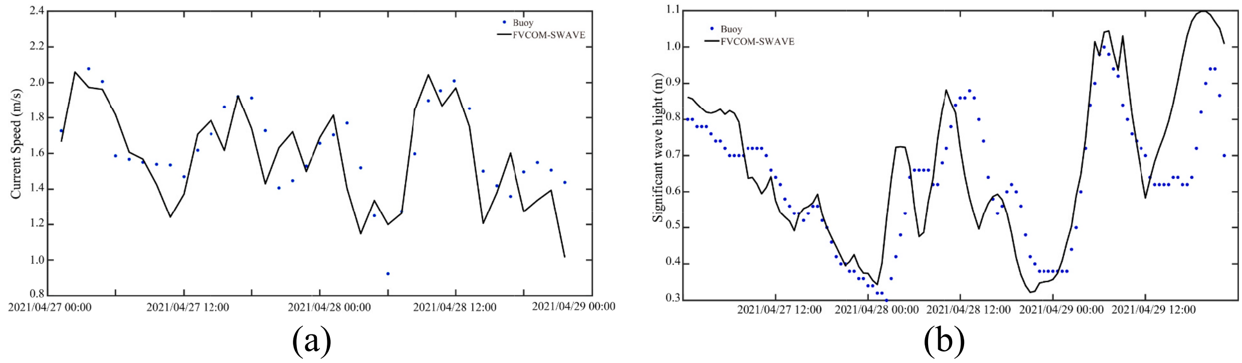

Fig. 6. Comparison between (a) current speed and (b) significant wave height by FVCOM-SWAVE model with the measurements from a buoy.

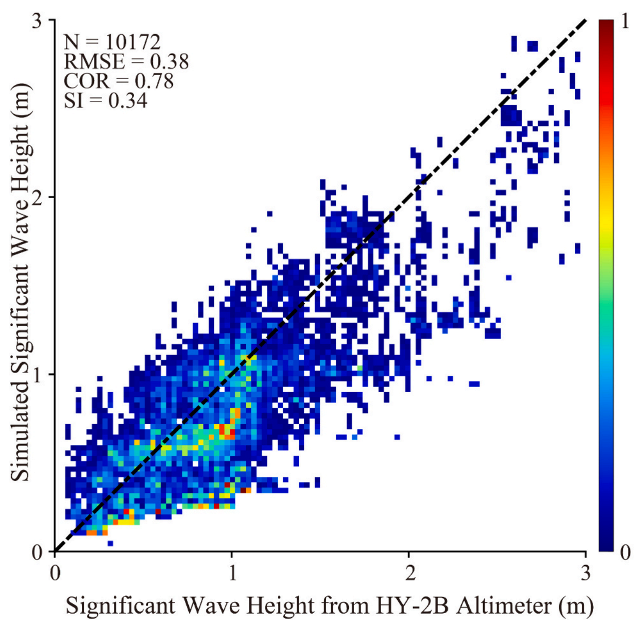

Fig. 7. Validation of simulated SWHs against measurements from the HY-2B altimeter from 15 April to 25 May 2021.

simulation is studied. Finally, an error analysis is conducted.

4.1. Validation of simulations using FVCOM-SWAVE

The HYCOM dataset is used in FVCOM-SWAVE model, therefore, we conduct qualitative analysis on HYCOM and simulation results. As an example, a current speed map by the coupled FVCOM-SWAVE model at 12:00 UTC on 3 May 2021 and the corresponding HYCOM current map is presented in Fig. 5a and b, respectively. In general, the pattern of the simulated current is consistent with that of the HYCOM data. Furthermore, it is necessary to figure out that the spatial resolution of the simulation results obtained using the FVCOM-SWAVE model can reach 1 km, which is higher than the 8 km resolution of the HYCOM field. The time series of the simulations (i.e., current speed and SWH) from the FVCOM-SWAVE model and the measurements from an available buoy were shown in Fig. 6. It was found that the variation in the simulation by FVCOM-SWAVE was consist with that from buoy. A comparison of the collocated SWHs from the modeling results and the observations from the HY-2B altimeter from 15 April to 25 May 2021 is shown in Fig. 7, in which the samples are grouped into a 0.1 m bin. Statistical analysis yielded a root mean square error (RMSE) of 0.38 m, a correlation coefficient (COR) of 0.78, and a scatter index (SI) of 0.34. Collectively, the model-simulated currents and waves by the FVCOM-SWAVE model in

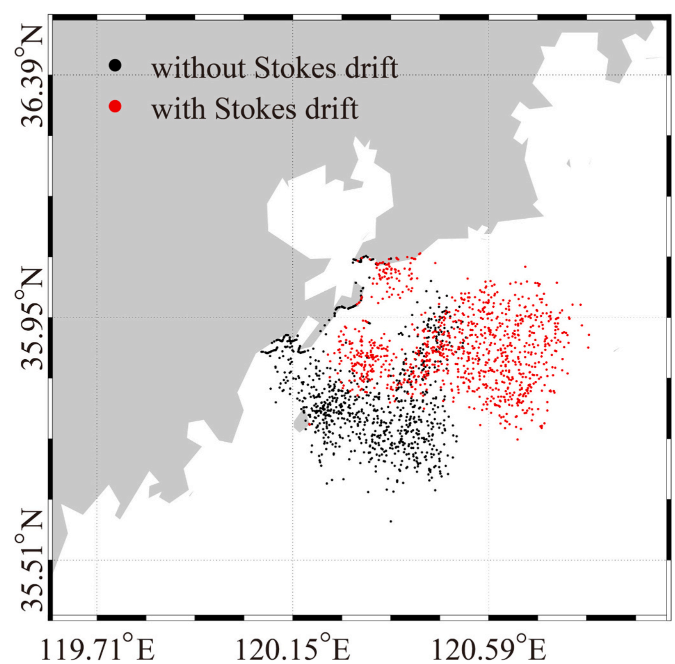

Fig. 8. Simulated oil spill map at 09:00 UTC on 20 May 2021.Noted that the red and black dots represent the results with and without Stokes drift, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

this study are reliable.

4.2. Results

As in a previous study (Qiao et al., 2019), the oil spill during the Symphony wheel accident from 17 April to 19 May 2021 was hindcast by using the particle-tracing method, in which the ERA-5 wind and FVCOM-SWAVE current were utilized as the forcing fields. In particular, the Stokes drift estimated using the simulated waves was tested in the hindcasting procedure so as to investigate the performance of waves on the oil spill simulation. Fig. 8 depicts the two types of simulations at 09:00 UTC on 20 May 2021. The red dots represent the results with Stokes drift, and black dots represent the results without Stokes drift. It can be clearly seen that there is a significant difference between them. In this case, we believe it is necessary to incorporate the role of Stokes drift in the oil spill simulation.

In addition, the oil spill detected using the GF-3 images was compared with the simulation results. Fig. 9 presents maps of the SAR-derived results (black dots) at 09:47 UTC on 19 May 2021. The black and red dots in Fig. 9a and b are the results without Stokes drift and the

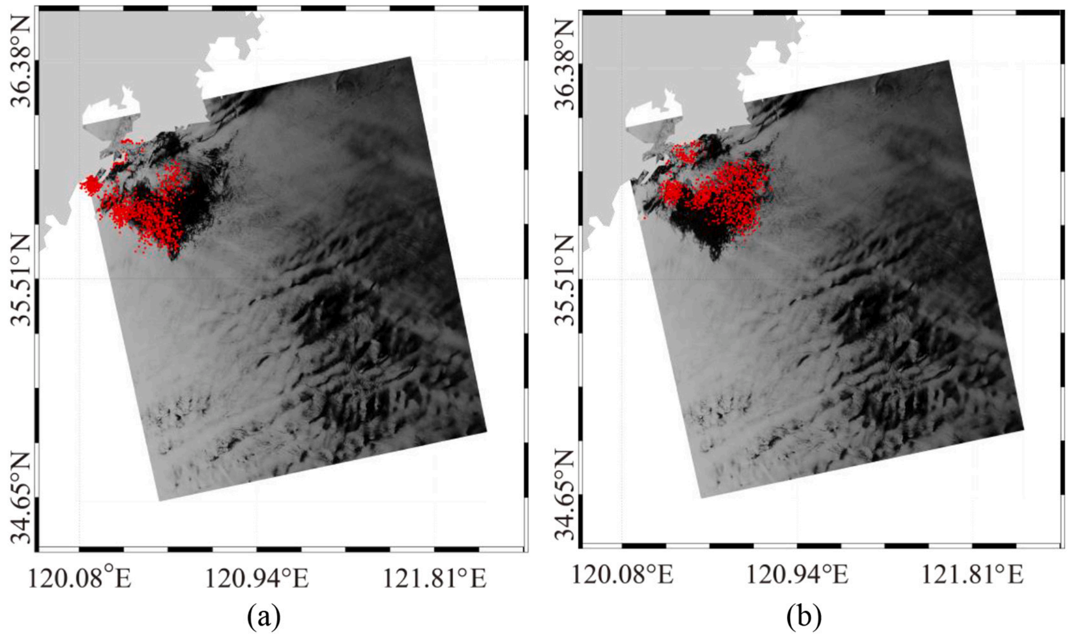

Fig. 9. Maps of the SAR-derived results at 09:47 UTC on 19 May 2021: (a) the results without Stokes drift (red dots) and (b) the results with Stokes drift (red dots). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Oil area of SAR-derived results and model-simulated oil spills.

Imaging Time	SAR (km^2)	Simulation without Stokes drift (km^2)	Simulation with Stokes drift (km^2)
2021-05-02 09:50 UTC	772.53	826.74	823.04
2021-05-04 10:07 UTC	165.10	1232.10	1408.29
2021-05-09 09:58 UTC	19.71	1032.50	1101.50
2021-05-19 09:47 UTC	1411.99	1485.91	1356.54

results with Stokes drift, respectively. The difference between the oil spill simulations without Stokes drift and the SAR-derived results can be clearly seen, and the simulations with Stokes drift have a better performance.

4.3. Discussion

In this section, the difference in the oil spill coverage area between the SAR-derived results from four images and the model-simulated oil spill (Table 3) is analyzed. Because the oil spills were not completely covered in the GF-3 images acquired on 4 and 9 May 2021, there is a large difference between the SAR-derived results and the simulation results (Fig. 10). It can be clearly seen that the spatial coverage of the model-simulated oil spill with Stokes drift is close to the SAR-derived results, i.e., a − 73.92 km^2 bias (SAR minus simulations) without Stokes drift and a 55.45 km^2 bias with Stokes drift on 19 May 2021. However, this behavior is not apparent for the image on 2 May 2021, which was acquired at initial stage of oil spills. The geographic location of the center point of the SAR-derived results for 19 May 2021 is (35.80°N, 120.58°E). Compared with the model-simulated oil spill, the bias with Stokes drift (geographic location of (35.74°N, 120.36°E)) is 8.18 km, which is less than the 12.95 km bias that is achieved without Stokes drift (geographic location of (35.79°N, 120.49°E)). In order to analyze the Stokes drift effect on the oil spill simulation, the distance of oil spills (SAR-derived results minus the model simulations) versus the ratio of Stokes drift to total current speed was conducted in Fig. 11. In general, the bias with the Stokes drift is low than the result without the Stokes drift, indicating that accuracy with Stokes drift is significantly higher than that without Stokes drift. With the ratio of Stokes drift increasing, the distance of oil spills without Stokes drift greatly oscillates, however, this behavior is improved with Stokes drift. In this sense, the accuracy of the model-simulated oil spill is improved by including the Stokes drift. The limitation of this study is limited SAR images for validation, thus, more SAR images is anticipated to be used in the future work.

5. Conclusion

At present, oil spills are conveniently predictable using a numeric model based on oil particle-tracing method. Although sea surface winds and nearshore currents (especially tides) are the main driving forces in a prediction model, the impact of the Stokes drift induced by waves is worthy of further study. Therefore, in this study, this issue was clarified using the Symphony wheel accident in the Qingdao coastal waters as a case study.

The currents and waves were synchronously simulated by using a coupled numerical model, that is the FVCOM-SWAVE model, on March–May 2021, and the ERA-5 wind was utilized as the forcing field and the open boundary conditions included sea surface temperature, salinity, current, and water level collected from open-accessed HYCOM dataset. In addition, four GF-3 images were acquired from 2 to 19 May 2021. Validation of the model-simulated SWHs against matchups from the HY-2B altimeter yielded an RMSE of 0.38 m, a COR of 0.78, and an SI of 0.34, demonstrating that the simulated by the FVCOM-SWAVE model are trustworthy for our work. Utilizing the ERA-5 wind, model-simulated currents, and waves, the oil spill was hindcast using the oil particle-tracing method. In addition, the oil spill was identified using the SAR images and a CFAR algorithm. At initial stage of oil spills such as the image on 2 May 2021, the difference in simulation results with or without Stokes drift is very small. Two images on 4 and 9 May 2021 only captured parts of oil spills, causing the large deviation between the SAR-derived results from and simulated oil spills. As for the image on 19 May 2021, the spatial coverage of the model-simulated oil spill without Stokes drift (i.e., a − 73.92 km^2 bias) is close to the SAR-derived results,

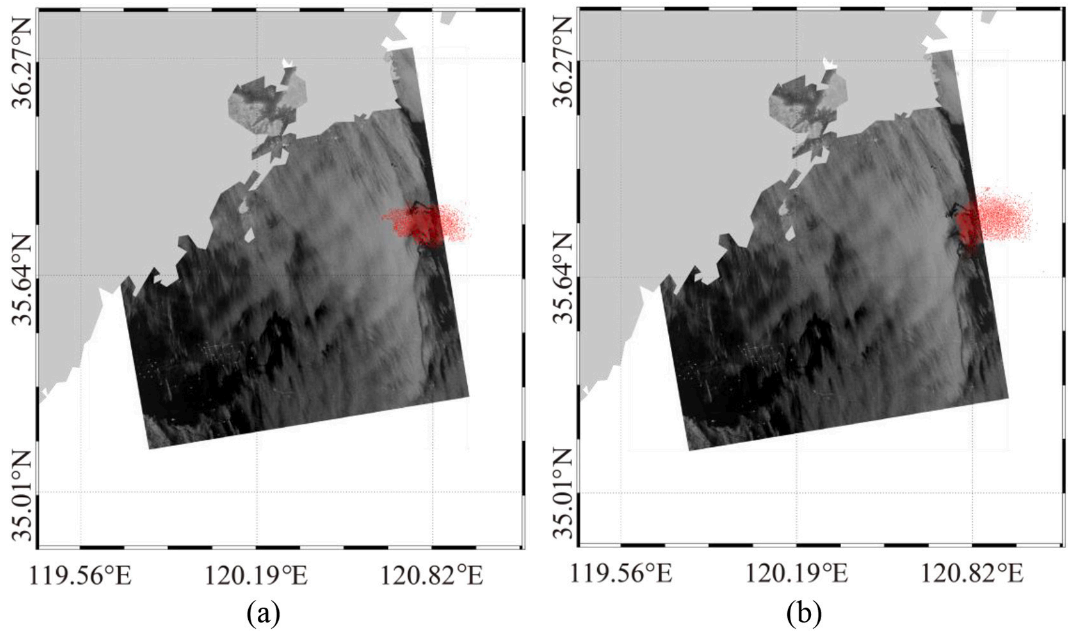

Fig. 10. Maps of SAR-derived results at 10:07 UTC on 4 May 2021: (a) the results without Stokes drift (red dots) and (b) the results with Stokes drift (red dots). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

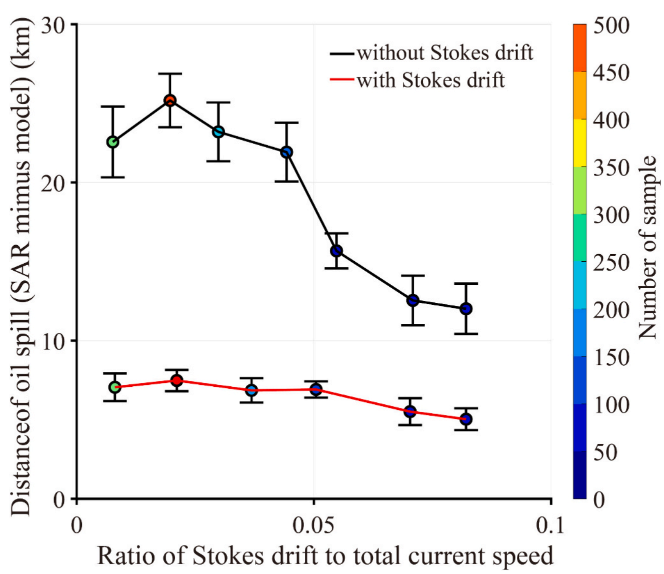

Fig. 11. Distance of oil spills (SAR-derived results minus the model simulations) with respect to the proportion of stokes drift in the surface current.

which is worse a 55.45 km^2 bias with Stokes drift. Similar result is also being discovered as comparing the geographic location of the center point of the SAR-derived results with those from simulated oil spills, i.e., 8.18 km bias achieved with Stokes drift and a 12.95 km bias achieved without Stokes drift. With the ratio of Stokes drift increasing, the significant oscillation existing in the distance of oil spills (SAR-derived results minus the model simulations) without Stokes drift is improved as including Stokes drift. Under this circumstance, it is believed that Stokes drift needs to be included in the prediction of oil spills.

As a matter of fact, sea surface wind is the major driver of oceanic dynamic processes. In the near future, numeric coupling model, i.e., Weather Research and Forecasting (WRF) and FVCOM-SWAVE, will be applied for atmospheric-oceanic dynamics simulation which can future improve the oil spill prediction.

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CRediT authorship contribution statement

Jiale Chen: Project administration, Investigation, Data curation. **Song Hu:** Writing – original draft, Methodology. **Yiqiu Yang:** Validation, Resources, Methodology, Investigation. **Xingwei Jiang:** Supervision, Funding acquisition. **Wei Shen:** Writing – review & editing, Methodology. **Huan Li:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Forecasts (ECMWF) reanalysis (ERA-5) wind field and the water depth from the General Bathymetric Chart of the Oceans (GEBCO) are the forcing field and bottom boundary in the numerical modeling.

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